

2 Entropy-informed feedback in autism neuromodulation 3 research: a falsifiable neural-mass benchmark

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11 **Abstract**

12 Person-specific closed-loop neuromodulation is a plausible research direction in autism, but neither an
13 autism-specific entropy target nor the added value of entropy feedback has been established. We
14 implemented an exploratory Python benchmark that separates neural dynamics, observation, feedback and
15 independent task evaluation. Sixteen coupled excitatory–inhibitory neural masses represented each of 24
16 synthetic individuals, without assigning an autism diagnosis to model parameters. Separately calibrated
17 controllers were evaluated across three random seeds, six nominal strategies and four robustness conditions,
18 yielding 1,296 evaluation runs. Adding permutation-entropy feedback to rate feedback reduced continuous
19 task-tracking mean squared error by 4.53 units (95% bootstrap interval across synthetic individuals, −6.54 to
20 −2.83 for combined minus rate feedback), while increasing control energy by 0.0175 model units. Binary
21 task accuracy remained approximately 50%, compared with 80.3% in a held-out unperturbed reference
22 session. Increased measurement noise reversed the incremental tracking advantage, and a two-second delay
23 made it uncertain. Calibrated fixed input had lower mean tracking error than either adaptive strategy.
24 Bounded covariance plasticity produced weight changes whose washout followed the imposed relaxation
25 rule. These results demonstrate conditional control performance and failure modes in a specified synthetic
26 system, not efficacy or durable therapeutic plasticity in autistic people. The contribution is a reproducible
27 test of whether complexity adds useful information beyond simpler feedback, accompanied by an interactive
28 anatomical display and an explicit translational research agenda. Clinical development requires reliable
29 individual measurements, realistic observation and actuation models, and benefits defined with autistic
30 participants.

31 **1 Introduction**

32 The appeal of combining a bidirectional brain–computer interface with the entropic brain framework is that
33 a system could measure changing neural dynamics, intervene when needed, and support useful learning.
34 However, three distinct propositions are often compressed into that narrative: that a neural statistic differs in
35 autism; that an intervention can alter that statistic; and that altering it improves an outcome valued by a
36 particular person. Evidence for one proposition does not establish the others. We therefore treat entropy-
37 informed feedback as an engineering hypothesis requiring an independent outcome, rather than as an
38 established therapy.

39 Autism heterogeneity makes a universal connectivity correction difficult to justify. In a 32-site analysis of
40 1,824 people, Ilioska et al. (2026) found that extreme deviations in individual functional connections rarely
41 overlapped between participants, although convergence increased at regional and network scales. This
42 combination supports multiscale individual characterization without implying an absence of shared
43 structure. A normative deviation is a statistical description; it does not, by itself, identify dysfunction, a
44 desirable reference state or an indication for intervention. Participatory research is necessary to determine
45 which difficulties merit support and which outcomes matter (Fletcher-Watson et al., 2019).

Neural complexity also depends on what is measured. Tenev et al. (2025) reported higher generalized EEG entropy but lower Lempel–Ziv complexity in autistic children relative to a comparison group. Fu et al. (2025) found a localized increase in sample entropy during sleep-state functional MRI, alongside directional differences in information transfer. Neither result establishes a global deficit to be corrected by increasing entropy. Estimator, temporal scale, recording modality, developmental stage and context must remain explicit.

The entropic brain theory concerns relationships between neural dynamics and conscious states (Carhart-Harris et al., 2014; Carhart-Harris, 2026). Recent work adds context to a simple disorder interpretation. Stoliker et al. (2026) found that psychedelic brain trajectories in 62 adults aligned with sensory conditions rather than becoming indiscriminately disorganized. Lyons et al. (2026) reported relationships between acute signal complexity and later wellbeing after first psilocybin use in a small exploratory adult study; these associations neither establish causal mediation nor demonstrate an autism treatment. Mechanistic whole-brain modeling links serotonergic effects to circuit and network properties rather than equating a drug with an arbitrary noise source (Herzog et al., 2023). These studies motivate measuring dynamics and context separately. They do not supply the therapeutic objective used here.

The clinical stimulation literature is mixed and protocol-specific. Two sham-controlled frontal theta-burst trials did not establish superiority on their primary autism symptom outcomes (Ni et al., 2023, 2024), consistent with uncertainty in a network meta-analysis of randomized noninvasive stimulation studies (Chen et al., 2024). A subsequent multicenter trial randomized 200 children to accelerated motor-cortex stimulation or sham and reported improved social-communication scores through one month (Tan et al., 2026). This positive trial updates an exclusively negative narrative, but it tested neither entropy targeting nor feedback control. Its protocol, population and follow-up cannot be generalized to an unspecified bidirectional device.

Bidirectional interfaces and adaptive stimulation provide complementary engineering precedents. Tactile feedback improved robotic-arm control in an implanted BCI study (Flesher et al., 2021), while individualized adaptive stimulation improved motor outcomes relative to optimized continuous stimulation in a four-person Parkinson’s disease feasibility trial (Oehrns et al., 2024). A small autism BCI study demonstrated feasibility of a social-attention training approach rather than definitive therapeutic efficacy (Amaral et al., 2018). None validates the proposed entropy mechanism in autism. Here, “bidirectional” refers narrowly to a modeled measurement–actuation loop; the benchmark does not implement a clinical sensory prosthesis.

The present work advances a supplied conceptual review and Julia prototype into an executable, falsifiable benchmark. We ask whether a calibrated entropy term improves an independent task readout beyond rate feedback, and whether any advantage survives delay, noise, stimulation artifact, yoked input and bounded plasticity. The model is intentionally synthetic: its purpose is to expose assumptions and counterexamples before patient fitting. It does not recreate autism, estimate an individual’s clinical state, or establish an optimal level of brain entropy.

2 Materials and methods

2.1 Study design and evidence update

The computational study was exploratory and was not preregistered. The combined-versus-rate tracking comparison was designated as the primary engineering contrast before the final benchmark execution; controller design and parameter ranges were developed through pilots. We report those choices and preserve a higher-noise pilot summary. A targeted evidence update used Consensus searches with subsequent paper-record retrieval, supplemented by primary publisher and PubMed records, through 7 September 2026. Search themes included autism neural complexity, sham-controlled stimulation, individual connectivity, adaptive BCI, entropic brain theory and neural-mass modeling. Publication metadata were checked against DOI records. This was a focused narrative update, not a systematic review, exhaustive search or meta-analysis. The evidence ledger documents source scope and unsupported extrapolations.

94 The original public repository was inspected at the source-file level. The audited Julia file had blob SHA
 95 f52f9b84294b6b6bb04a56218c2a97471867c8c5. Its random-walk demonstration did not implement an
 96 entropy estimator or activity-dependent synaptic weights, and its classifier distinguished region identity
 97 rather than a functional task. The notebook contained recorded undefined-variable errors. The Python
 98 implementation is a new benchmark, not a line-by-line translation or a biological validation of that
 99 prototype. The full audit is supplied with the code.

100 2.2 Synthetic individuals and neural dynamics

101 Each individual comprised 16 Wilson–Cowan-type excitatory and inhibitory populations (Wilson and
 102 Cowan, 1972). We selected this scale for transparent controls and interactive execution. The neural-mass
 103 approach is consistent with reusable whole-brain simulation frameworks such as neurolib, although the
 104 present implementation does not depend on that package (Cakan et al., 2023). The two hemispheres and four
 105 paired modules were illustrative labels. No participant imaging, diagnostic label, receptor map or patient-
 106 specific parameter fit was used.

$$107 \quad \tau_E \, dE_i/dt = -E_i + S(a \, z_i)$$

$$108 \quad z_i = w_{EE,i} E_i - w_{EI,i} I_i + g \sum_j W_{ij} E_j + p - 2.5$$

$$109 \quad + q_i(t) + d(t) + b_i u(t) + \xi_i(t)$$

$$110 \quad \tau_I \, dI_i/dt = -I_i + S[a(10 E_i - 2 I_i - 3)], \quad S(x) = 1/(1 + \exp(-x)).$$

111 E and I are bounded normalized population activities, not calibrated firing rates in hertz. Time constants
 112 were 40 and 80 ms. Initial activities were 0.1. The noise process ξ was an Ornstein–Uhlenbeck current with
 113 150-ms time constant and stationary-scale parameter 0.04. Its stochastic forcing is a model component, not
 114 “entropy injection” or a pharmacological mechanism. Gain a, coupling g and baseline drive p varied
 115 between synthetic individuals; Table 1 specifies all ranges and fixed coefficients.

116 For each off-diagonal connection, an initial positive weight was proportional to $[0.15 + 0.60 M_{ij} + 0.30 H_{ij}]$
 117 times a lognormal factor with log-standard-deviation 0.30. M indicates the same paired module across either
 118 hemisphere; H indicates the same hemisphere. Rows were normalized to one and diagonal weights set to
 119 zero. A fixed broad actuator field b had values $[0.35, 0.35, 0.65, 0.65, 1, 1, 0.75, 0.75]$ in each hemisphere.
 120 This field defines an input distribution only. It is not a modeled electric field, coil position or implant
 121 configuration.

122 2.3 Task, perturbations and independent evaluation

123 Each 60-s run contained a balanced-in-expectation random binary input updated every 0.5 s. Input
 124 amplitude 0.6 entered four designated sensory populations. A scalar context disturbance was absent before
 125 15 s, equaled the individual disturbance parameter during 15–30 s, changed to -0.6 times that parameter
 126 during 30–45 s, and ended at 45 s. These manipulations test distribution shift in a dynamical system. They
 127 do not operationalize sensory overload, an autistic trait or a clinical episode.

128 An individual ridge readout used the 12 non-input populations to estimate the input with a 150-ms lag.
 129 Features were centered and scaled using a separate unperturbed calibration session; coefficients were fitted
 130 on seconds 10–45 with penalty 10. The readout and calibration statistics were held fixed during evaluation.
 131 A fresh unperturbed seed assessed baseline generalization. The readout accesses true latent excitatory states,
 132 giving a favorable observation assumption for task decoding; the feedback controller sees only a noisy
 133 population average. This asymmetry is deliberate and limits realism.

134 The primary engineering outcome was mean squared error (MSE) of the continuous readout during seconds
 135 15–45. Secondary outcomes included sign-classification accuracy, mean squared deviation from reference
 136 activity, permutation entropy, spectral entropy, control energy and post-disturbance tracking error. The
 137 linear readout is unbounded: large MSE can result from gain or offset drift despite similar sign decisions.
 138 For that reason accuracy is reported alongside MSE. Neither metric measures social communication,
 139 wellbeing or therapeutic benefit. Washout readout error used seconds 47–60 to allow a short transition
 140 interval.

141 2.4 Observation and entropy estimation

142 The controller observed $y(t) = \text{mean}_i E_i(t) + \varepsilon(t) + A u(t) \sin(2\pi \cdot 37t)$, sampled at 100 Hz. Independent
 143 Gaussian sensor noise ε had nominal standard deviation 0.001. A was zero except in the stimulation-artifact
 144 condition, where it equaled 0.20. The 37-Hz term is an illustrative measurement contaminant, not a
 145 complete stimulation artifact model or EEG forward solution. No band-pass filter, source reconstruction or
 146 artifact rejection was applied.

147 Normalized permutation entropy was computed from ordinal patterns of dimension three, with lag two
 148 samples and one-second non-overlapping windows (Bandt and Pompe, 2002). Stable temporal ordering
 149 resolved equal values. With six possible ordinal patterns, $H = -\sum_k p_k \ln(p_k)/\ln(6)$, which ranges from zero to
 150 one. Each window contributes 96 overlapping embedded patterns. This short-window estimator has finite-
 151 sample bias and correlated observations; no claim of bias-free entropy estimation is made. An independent
 152 35-s baseline supplied the reference mean activity and mean window entropy. References therefore reflect
 153 the individual model's unperturbed task condition, not an empirically optimized wellbeing state.

154 To estimate how input changes the measured statistic, constant calibration inputs of +0.20 and -0.20 were
 155 applied in separate matched unperturbed simulations. Their mean entropy difference, divided by 0.40,
 156 defined an individual local slope s . If $|s|$ was below 0.01, the entropy correction abstained. Otherwise the
 157 correction $v = \text{clip}[(H^* - H)/s, -0.35, 0.35]$ used the sign and scale of the measured response. This
 158 derivative is local and noisy; it does not establish global controllability or causal specificity of entropy.

159 2.5 Feedback strategies and comparators

160 Controllers updated once per second using the latest available one-second measurement window. Rate
 161 feedback proposed $u = 4(r^* - r)$. Entropy-only feedback proposed 0.5 times the preceding input plus 0.5v.
 162 Combined feedback proposed $4(r^* - r) + 0.3v$. All proposals passed through a slew limit of 0.15 per update
 163 and an absolute amplitude cap of 0.60. Proposals were zero before 10 s and from 45 s onward, with the same
 164 slew constraint retained during turn-off. These are dimensionless numerical limits and cannot be converted
 165 into a safe human stimulation dose.

166 The no-input comparator shared the task and random streams. The fixed-input comparator selected the
 167 lowest-MSE value among seven equally spaced candidates from -0.60 to +0.60 on an independent
 168 perturbed calibration seed. Thus fixed input had access to the disturbance distribution during calibration,
 169 whereas adaptive rules were not optimized on the evaluation outcomes. It is a useful, relatively strong open-
 170 loop comparator, but this design does not compare equally optimized controller families. A yoked
 171 comparator replayed the combined controller's input from the next individual in a cyclic ordering,
 172 preserving aggregate input energy while removing current-person feedback. It controls aggregate exposure
 173 rather than energy matching within every individual.

174 2.6 Robustness, plasticity and entropy diagnostics

175 The control architecture is summarized in Figure 1. The nominal experiment evaluated six strategies across
 176 24 individuals and three seeds. Four additional conditions each evaluated no input, rate feedback and
 177 combined feedback: two-second extra observation delay; sensor-noise standard deviation 0.05; artifact scale
 178 0.20; and activity-dependent weight adaptation. These conditions were studied separately rather than as a
 179 full factorial experiment. The moderate-noise pilot used standard deviation 0.01 and yielded weak entropy
 180 calibration in much of the ensemble. The final low-noise condition deliberately tests a favorable
 181 measurement case, not expected EEG performance.

182 In the plasticity condition, off-diagonal weights followed a bounded covariance rule with a two-second
 183 running activity mean, coefficient $\eta = 0.15$ and relaxation time 20 s:

$$184 \quad dW_{ij}/dt = \eta(E_i - \bar{E}_i)(E_j - \bar{E}_j) - (W_{ij} - W_{ij}^0)/20.$$

185 The covariance term ended at 45 s; relaxation continued. Weights were constrained to $[0.25 W^0, 2 W^0]$, with
 186 zero diagonal and no subsequent row normalization. This phenomenological rule is neither spike-timing-
 187 dependent plasticity nor a validated molecular mechanism. Relative Frobenius weight change quantified

188 adaptation and washout. Residual changes cannot establish durability because retention is largely specified
189 by the imposed decay constant.

190 Exploratory gain sweeps covered 0.9–2.4 at 11 levels in eight individuals, without imposing an inverted-U
191 relationship between complexity and performance. Forty Fourier phase surrogates per individual preserved
192 the magnitude spectrum of a 30-s observed record. Original and surrogate permutation entropy were
193 compared descriptively. Whole-record surrogate entropy and mean one-second online entropy use different
194 windows and are not numerically interchangeable. These diagnostics examine dependence on dynamics and
195 temporal ordering; they do not identify an autism biomarker or a universal optimal entropy.

196 **2.7 Computation, uncertainty and verification**

197 Forward Euler integration used a 2-ms step and recorded every 10 ms. A seeded 32-bit linear congruential
198 generator with Box–Muller Gaussian conversion enabled a matched JavaScript implementation. At a fixed
199 step size, arms shared task and stochastic streams; individual parameters used NumPy’s separately seeded
200 generator. Parameter seeds were 4100 plus individual index, calibration seeds were 710 plus 11 times that
201 index, and evaluation seeds were 100, 200 and 300. Supplementary code specifies all streams, including
202 validation and diagnostic seeds.

203 Replicates were averaged within each synthetic individual. Mean contrasts and 95% percentile intervals
204 used 4,000 bootstrap resamples of individuals with seed 809. These intervals describe this sampled
205 parameter ensemble, not patients, parameter posterior uncertainty, or all plausible neural models. Secondary
206 contrasts were exploratory and were not adjusted for multiple comparisons; no confirmatory P values are
207 reported. Results were retained regardless of direction.

208 Thirteen software and numerical checks assessed reproducibility, estimator invariance, bounded states,
209 constrained input, matched task streams, abstention, yoked energy, data integrity and plasticity decay. An
210 independent high-accuracy adaptive ODE solver checked the production integrator during a deterministic
211 0.5-s interval; decreasing the Euler step from 2 to 1 to 0.5 ms reduced maximum activity error. A separate
212 stochastic step-size sensitivity file is provided; using the same seed at different steps does not create a
213 shared Brownian path, so that comparison is not a pathwise convergence proof. Code, raw evaluation rows,
214 calibration records, figures and an executable notebook accompany the manuscript, consistent with
215 reproducible computational practice (Sandve et al., 2013).

216 **2.8 Interactive anatomical representation**

217 The browser interface renders the fsaverage5 cortical template distributed through Nilearn from FreeSurfer,
218 with folded and inflated surfaces. The 16 synthetic states color nearby vertices and the display shows
219 illustrative network edges. Anatomical geometry, synthetic regional labels and simulated dynamics remain
220 separate: the surface is not an autistic participant’s brain or an empirical activation map. A Web Worker
221 executes the JavaScript port, while Python-generated benchmark data provide fixed reference results. A
222 seed-matched example reproduced Python population trajectories to the six-decimal precision of the display
223 data. The notebook is the reproducible analysis record; the browser is an exploratory interface, not a clinical
224 control application.

225 **3 Results**

226 **3.1 Calibration and nominal controller performance**

227 The held-out unperturbed reference achieved 80.27% task accuracy (95% synthetic-ensemble interval 78.48
228 –82.13%) and MSE 0.706 (0.663–0.748). All estimated entropy slopes were negative in the chosen
229 operating range, spanning –0.1356 to –0.0059; two of 24 fell below the abstention threshold. Thus the
230 model does not encode the premise that stimulation always increases entropy.

231 Under context disturbance, mean MSE was 124.41 without input, 69.94 with calibrated fixed input, 81.36
232 with rate feedback, 97.80 with entropy feedback, 76.83 with combined feedback, and 126.77 with yoked
233 replay (Figure 2; Table 2). Combined minus rate feedback was –4.53 MSE units (–6.54 to –2.83). The

corresponding difference in control energy was +0.0175 (+0.0147 to +0.0202). Combined feedback also had lower MSE than yoked replay by 49.95 units (20.14–86.17 lower), indicating dependence on current-individual state under this design. Aggregate energy was identical for combined and yoked input.

The MSE advantage did not translate into useful sign discrimination. Mean accuracy was 50.04% with rate feedback and 50.03% with combined feedback; their difference was −0.008 percentage points (−0.020 to approximately zero). The calibrated fixed comparator reached 52.58% and had the lowest mean MSE, although its calibration access differed. These findings support partial correction of a disturbed continuous readout, not restoration of the original task representation or superiority of adaptive control in general.

3.2 Measurement conditions changed the incremental result

With two seconds of extra observation delay, combined minus rate MSE was −0.77 (−3.00 to +1.57). With sensor-noise standard deviation 0.05, it was +8.28 (+4.60 to +12.65), reversing the nominal advantage. With stimulation artifact, the contrast was +3.52 (−0.11 to +7.56), with uncertainty spanning zero (Figure 3). The results show that nominal performance cannot be extrapolated across observation conditions. They do not identify a clinically acceptable noise threshold or quantify performance with a real recording device.

3.3 Plasticity and diagnostic checks

The bounded-plasticity condition retained a combined-versus-rate MSE difference of −4.60 (−6.51 to −3.01). In the validation example, weight change at the end of washout was 0.47235 times its value at 45 s, matching the prescribed exponential relaxation. This is numerical evidence that the implemented decay rule behaves as specified, not evidence of long-term learning. The exploratory gain and phase-surrogate results illustrate that entropy depends on operating point and temporal organization (Figure 4); they provide no patient-derived entropy optimum.

All 13 verification checks passed. Against the independent deterministic solver, maximum excitatory-state errors were 1.92×10^{-5} , 1.01×10^{-5} and 5.14×10^{-6} at steps of 2, 1 and 0.5 ms, respectively. These checks support implementation integrity within their tested scope. They do not resolve biological validity, stochastic convergence across all regimes, or model identifiability.

4 Discussion

4.1 What the benchmark adds

The principal advance is a falsifiable separation between controlling a measured statistic and preserving independent information. In the favorable nominal condition, an entropy term improved continuous tracking beyond rate feedback, but the advantage required greater input energy and did not rescue classification accuracy. A simpler calibrated fixed input had lower mean error. These outcomes narrow the hypothesis: entropy may contain locally useful control information in a particular operating regime, yet “adaptive,” “complex” and “therapeutic” are not interchangeable descriptions.

The noise reversal is especially informative because entropy can be changed by the observation process itself. Feedback then acts on a mixture of latent dynamics, task structure and measurement contamination. A future system would need to estimate which part of a complexity change is interpretable and when its uncertainty warrants abstention. The present slope threshold offers only a minimal example. It does not address session-to-session calibration drift, nonstationarity, artifact rejection or state-estimation uncertainty. Those omissions are testable engineering gaps rather than evidence against all closed-loop approaches.

No causal benefit of entropy itself has been established. The combined controller changes input, rate and higher-order signal statistics together. Rate and entropy may reflect overlapping information; lower MSE could follow changed operating point rather than a distinct complexity mechanism. An energy-matched, optimized rate controller, phase-scrambled feedback, multiple estimators, explicit measurement models and mediation-resistant task controls are stronger next comparisons. The yoked condition addresses one alternative explanation but cannot substitute for these ablations.

279 4.2 Updating the therapeutic rationale

280 The recent literature supports a more precise rationale than either universal normalization or dismissal of
 281 stimulation research. Multiscale heterogeneity argues for individual characterization (Ilioska et al., 2026),
 282 while context-dependent psychedelic dynamics motivate distinguishing structured variability from noise
 283 (Stoliker et al., 2026). Neither observation defines a clinical target. The positive motor-cortex trial and
 284 negative frontal trials show that intervention claims must retain their population, target and outcome
 285 boundaries (Ni et al., 2023, 2024; Tan et al., 2026). The current model includes no implementation of those
 286 protocols and cannot explain their treatment effects.

287 Person-specific references should therefore be selected through repeated measurement and outcomes the
 288 participant values. Returning to an individual's own baseline may be a useful engineering control, but it is
 289 not automatically desirable if that baseline is distressing or constraining. Equally, being statistically atypical
 290 is not sufficient reason to intervene. The translational objective is assistance with consented difficulties,
 291 such as a personally prioritized communication or sensory goal, while preserving autonomy and avoiding
 292 pressure to suppress harmless autistic characteristics. Meaningful participation should shape the question
 293 and stopping decisions, not merely recruitment (Fletcher-Watson et al., 2019).

294 4.3 A staged translational program

295 The next empirical step is a preregistered measurement study, designed with autistic adults and relevant
 296 clinical collaborators, before an efficacy trial. Candidate tasks and complexity estimators should be
 297 evaluated across repeated sessions, contexts and artifact conditions. The primary question would be whether
 298 a candidate statistic predicts a within-person, agreed functional outcome beyond simpler measures such as
 299 power, rate proxies and task history. Splitting data by session rather than adjacent windows would reduce
 300 leakage. Reliability, calibration transfer and the ability to detect uncertainty should determine whether the
 301 feedback hypothesis proceeds.

302 Model development should then replace the synthetic coupling, population-average observation and broad
 303 actuator with appropriate data and forward models. ABIDE resources are valuable for autism imaging
 304 research, but their available modalities must be inspected rather than assuming a ready-made individual
 305 diffusion connectome for every participant (Di Martino et al., 2014, 2017). Empirical fits would require
 306 held-out validation, sensitivity to alternative model families, identifiable parameters and comparison with
 307 simpler null models. A realistic EEG observation model and input artifact mechanism are priorities because
 308 the present result changes direction under measurement noise.

309 Only after these steps should a clinically governed feasibility study test whether feedback adds benefit to a
 310 specified intervention. The protocol would need a justified modality, prospectively selected outcome,
 311 credible masking, fixed and yoked comparators, blinded assessment and follow-up sufficient for the
 312 proposed duration of benefit. Success should require a participant-valued improvement beyond simpler
 313 control, acceptable burden, reproducible measurement and no unacceptable adverse effects. No stimulation
 314 dose, implant recommendation, psychedelic regimen or treatment eligibility criterion follows from this
 315 benchmark. Human study design requires appropriate expertise, ethics review and participant consent; none
 316 has been obtained or claimed here.

317 4.4 Limitations

318 This work is a small exploratory computational study with a deliberately favorable nominal sensor.
 319 Parameter ranges were selected during development and do not represent a measured autism distribution.
 320 Only 24 synthetic individuals and three evaluation seeds were used; bootstrap intervals do not incorporate
 321 structural model uncertainty. The fixed controller received a distinct calibration advantage, and adaptive
 322 gains were not globally optimized. The binary readout is a narrow engineering task, with unusually large
 323 errors under disturbance and privileged access to latent states. Its poor discrimination is a substantive
 324 negative result.

325 The estimator uses short windows, one embedding dimension and one lag. Delays, artifacts and plasticity
 326 were tested separately, with no full joint stress analysis. Coupling has no conduction delays, adaptation is

phenomenological, and neither the actuator nor the observer is physiologically fitted. Real cortical geometry in the interface does not increase biological validity. The local deterministic numerical check is not a proof for the complete stochastic feedback system. Finally, the literature update was targeted and non-exhaustive. These limits preclude clinical efficacy claims, diagnostic use, assertions of durable rewiring and general conclusions that more or less entropy is beneficial.

5 Conclusion

Entropy-informed feedback deserves testing as an incremental source of control information, not adoption as a therapeutic objective. In this synthetic benchmark it improved one continuous measure under clean observations, failed to restore task discrimination and became disadvantageous with greater sensor noise. The accompanying code and interface make those boundaries reproducible and inspectable. Advancing the autism research program requires reliable individual measurements and prospectively defined, participant-valued outcomes before therapeutic claims can be assessed.

Data and code availability

All data in this study were generated computationally. The accompanying BBCI_Research_Package contains model source, scripts, calibration data, all 1,296 evaluation rows, summary results, verification output, figure source and an executed notebook. The original prototype is available at <https://github.com/ProCO-RPHopkins/BBCI-Entropic-Plasticity-Simulation>. The new package has not been deposited in a public archival repository and has no assigned DOI; a versioned public deposit should be completed before journal submission. No external patient data were accessed.

Ethics statement

This work used synthetic simulations and publicly distributed template anatomy. It involved no recruitment, patient records, human stimulation or animal experiments. No institutional ethics approval or human-study registration is claimed. Any future participant study is a separate project requiring appropriate review and consent.

Author contributions

Ryan Hopkins originated the research question and supplied the prior manuscript and Julia prototype. The present draft, implementation and analysis were developed with AI assistance. Final CRediT roles, authorship eligibility, independent scientific verification and approval of the complete manuscript remain to be confirmed by the human author before submission.

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OpenAI ChatGPT/Codex assisted with literature discovery, synthesis, Python and JavaScript implementation, computational analysis and manuscript drafting in September 2026. The exact underlying model version was not exposed in the working interface and must be added to the submission disclosure if available. Consensus supported scholarly discovery; publisher and DOI records supported source checking. An earlier Claude-assisted manuscript supplied by the author informed the starting question; its model

version was not specified. All manuscript figures were generated from explicit code and simulation data, without generative-image processing. AI systems are not listed as authors. The human author must verify scientific claims, citations, code and declarations and accept responsibility before submission.

Supplementary material

The research package provides the executable notebook, parameter and calibration files, complete evaluation results, exploratory diagnostics, higher-noise pilot summary, original-model audit, evidence ledger and reproducibility instructions. Presentation materials contain explanatory artwork and are distinct from the data figures in this manuscript.

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- 435

436 Table 1. Model and analysis parameters. All inputs and activities are dimensionless unless a time unit is given. Ranges
 437 define synthetic variation, not autistic population estimates.

438

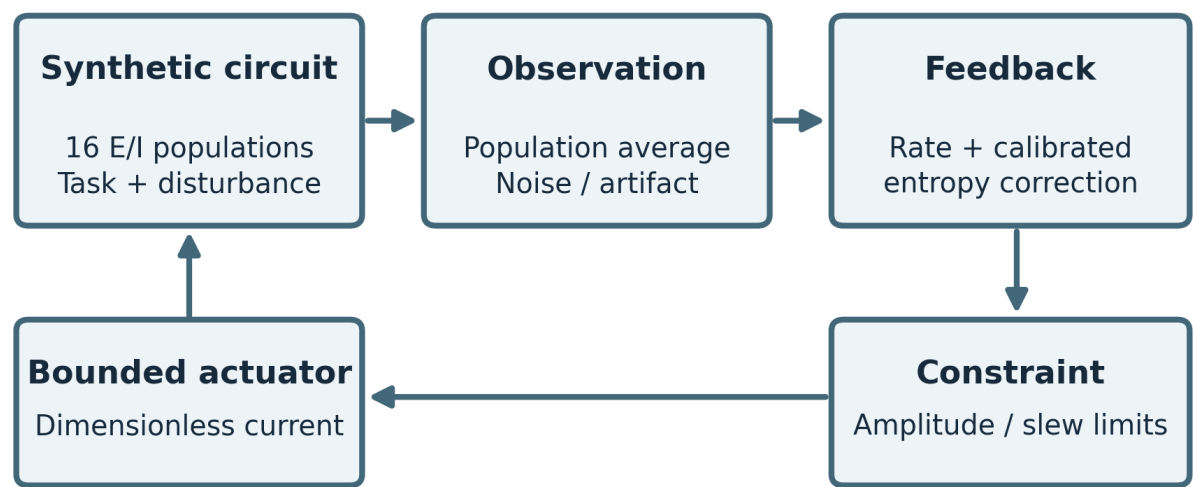
Component	Value / implementation
Ensemble	24 individuals; 16 E/I populations each; 3 evaluation seeds
Integration / observation	Euler 2 ms; output 100 Hz; 60 s per run
Time constants	Excitation 40 ms; inhibition 80 ms; current noise 150 ms
Gain / coupling / drive	Uniform 1.25–1.75 / 1.4–2.6 / 0.55–1.05
Local recurrent weights	wEE uniform 2.5–4.5; wEI uniform 9–11; I coefficients 10 and 2
Disturbance	Random sign $\times$ uniform 0.35–0.80; second context multiplies by -0.6
Intrinsic current noise	Ornstein–Uhlenbeck scale 0.04
Sensor noise / artifact	Nominal SD 0.001; stress SD 0.05; artifact $0.20 \times u \times \sin(2\pi \cdot 37t)$
Permutation entropy	Dimension 3; lag 20 ms; 1-s windows; normalization $\ln(6)$
Calibration	Separate reference; local entropy slope from ± 0.20 inputs
Entropy abstention / correction	$ \text{slope} < 0.01$; otherwise correction clipped to ± 0.35
Feedback gains	Rate: 4; combined: rate + $0.3 \times$ entropy correction
Actuator constraints	Absolute cap 0.60; slew 0.15 per 1-s update
Fixed comparator	7 candidates, -0.60 to $+0.60$, selected using an independent perturbed seed
Readout	12 non-input populations; ridge penalty 10; target lag 150 ms
Plasticity (optional)	$\eta = 0.15$; activity mean 2 s; relaxation 20 s; bounds $[0.25W^0, 2W^0]$
Uncertainty	4,000 bootstrap draws; seeds averaged within individuals

439

440

Table 2. Nominal engineering outcomes. Values are means with 95% percentile bootstrap intervals over 24 synthetic individuals after averaging three seeds. These are not clinical outcome estimates.

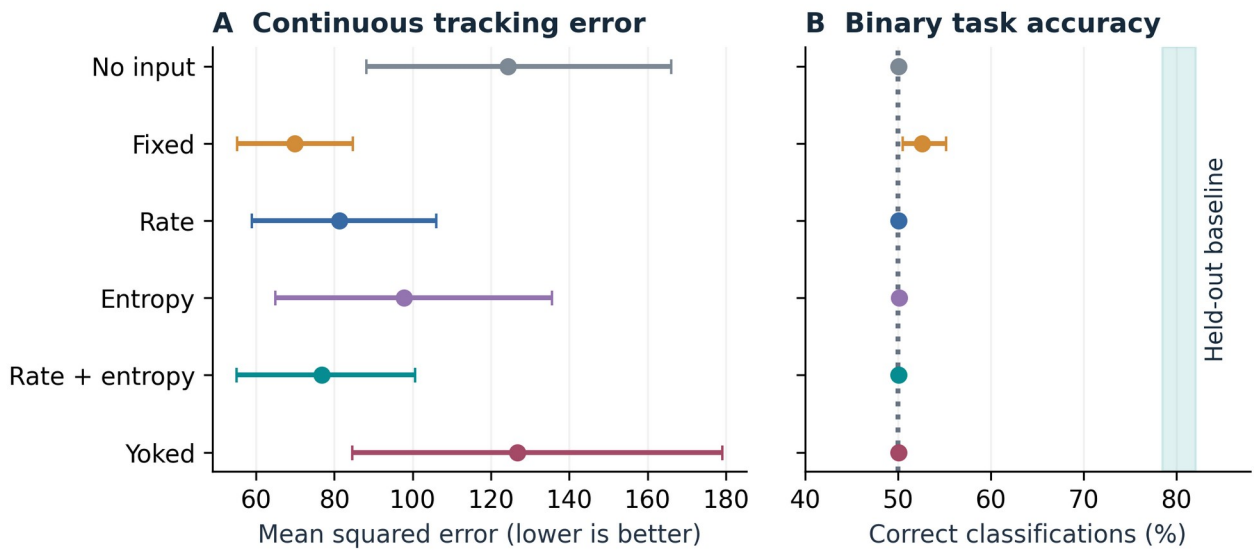
Strategy	Tracking MSE	Accuracy (%)	Input energy
No input	124.41 [88.14, 166.08]	50.04 [49.85, 50.23]	0.0000 [0.0000, 0.0000]
Fixed input	69.94 [55.26, 84.81]	52.58 [50.50, 55.21]	0.1917 [0.1367, 0.2483]
Rate feedback	81.36 [58.98, 106.10]	50.04 [49.85, 50.24]	0.0290 [0.0248, 0.0333]
Entropy feedback	97.80 [64.99, 135.56]	50.09 [49.87, 50.35]	0.0480 [0.0408, 0.0544]
Rate + entropy	76.83 [55.10, 100.63]	50.03 [49.84, 50.23]	0.0465 [0.0398, 0.0534]
Yoked replay	126.77 [84.61, 179.11]	50.06 [49.87, 50.28]	0.0465 [0.0399, 0.0532]



**Independent task readout evaluates success.
Entropy matching is not the outcome.**

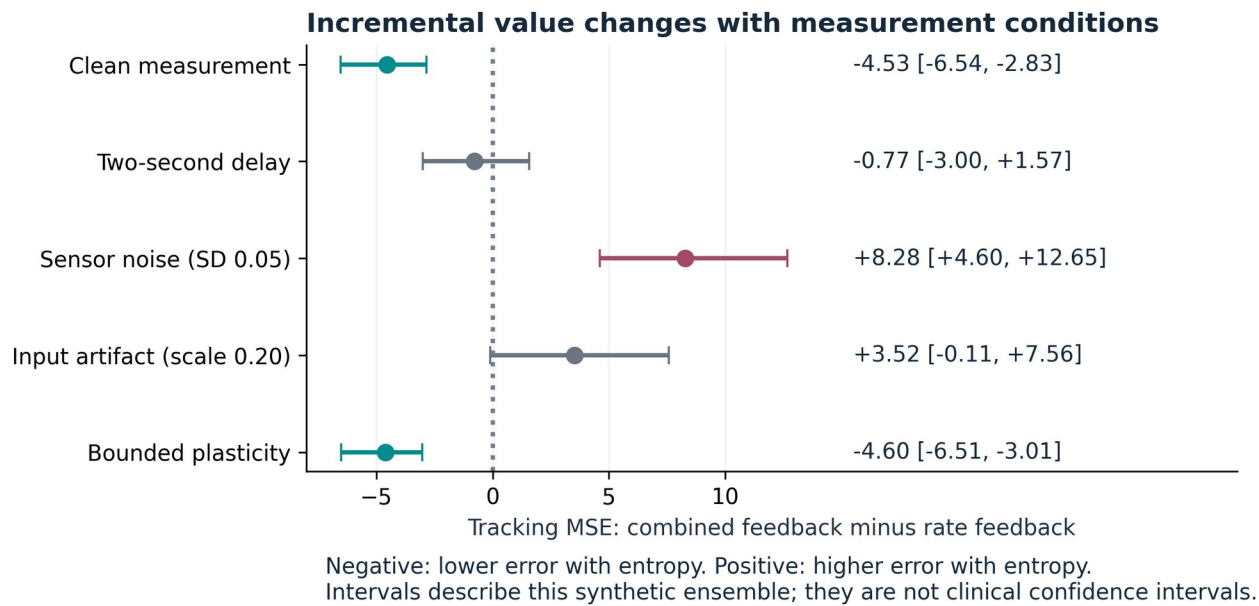
446
447 Figure 1. Measurement, feedback and independent evaluation are separated. The synthetic circuit generates an
448 observation; feedback uses reference errors; numerical constraints bound the returned input. A separately trained task
449 readout evaluates performance. Anatomy and therapeutic efficacy are not inferred from this architecture.

450



24 synthetic individuals $\times$ 3 seeds; intervals resample individuals (4,000 bootstrap draws).

Figure 2. Nominal controller outcomes. (A) Mean task-tracking squared error. (B) Binary task accuracy, with 50% shown as a chance reference and the held-out unperturbed accuracy interval shaded. Point estimates and intervals are based on 24 synthetic individuals, each averaged over three seeds. Combined feedback lowers tracking error relative to rate feedback without recovering useful sign discrimination. Fixed-input calibration access differs from that of adaptive rules.



458
459 Figure 3. Incremental tracking error from adding entropy feedback. Negative values favor combined feedback over rate
460 feedback; positive values favor rate feedback. Clean observations and optional plasticity have negative intervals, sensor
461 noise has a positive interval, and delay and artifact intervals span zero. The four stress conditions were tested
462 separately. Intervals describe synthetic ensemble variation.
463

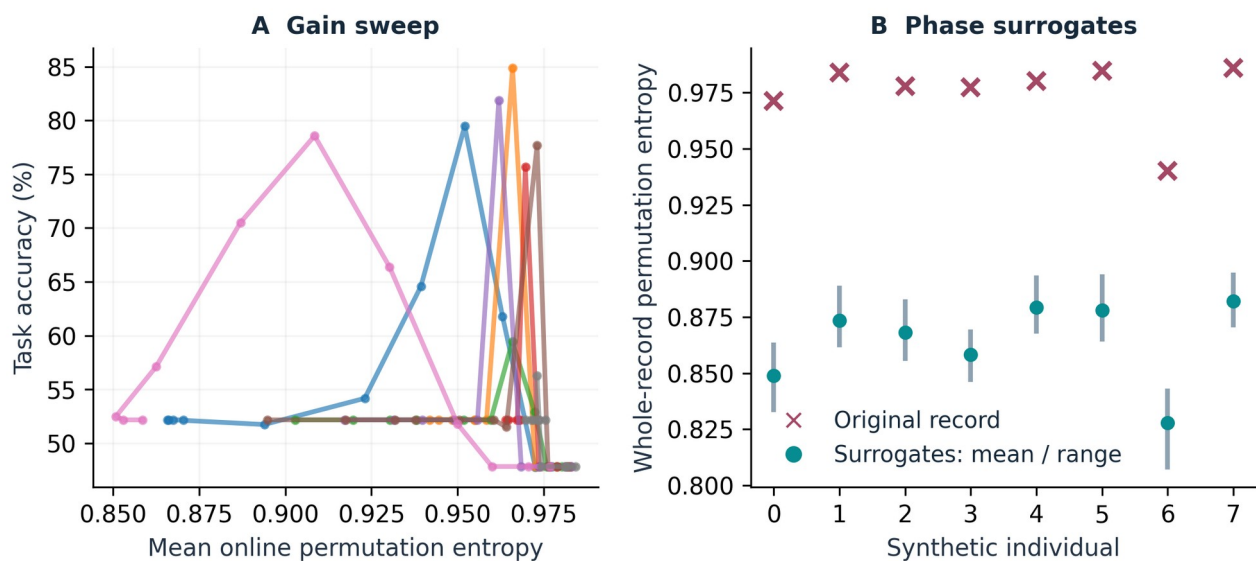


Figure 4. Exploratory entropy diagnostics. (A) Each line connects 11 gain settings for one of eight synthetic individuals. No optimal or inverted-U entropy curve was fitted. (B) Original whole-record permutation entropy is compared with the mean and range of 40 phase surrogates preserving Fourier magnitudes. Whole-record estimates differ from the mean short-window estimator used online. These descriptive diagnostics do not establish biomarker specificity.